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320413 (20)

BE (4th Semester)

Examination, Nov.-Dec., 2011

Branch : Civil

SURVEYING - II

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Answer all five questions. From all questions part (a) is compulsory. Out of parts (b), (c) & (d) answer any two carrying 7 marks. Assume any missing data suitably.

Unit-1

Q. 1. (a) Name different types of EDM instruments.

(b) What are the types of corrections to applied in base line ?

(c) What do you understand by phase of signal ? Derive the formula for reducing observation to correct the phase.

(d) Two triangulation station 'A' and 'B' are 60 apart and have elevation 240 m and 280 respectively. Find the minimum height signal required at 'B' so that the line of si may not pass near the ground than metres. The intervening ground may

assumed to have a uniform elevation of 200 metres. 7

Unit-2

2. (a) Define the 'most probable value'. 2

(b) Explain the method of correlates. 7

(c) Find the most probable values of angle A, B and C of triangle ABC from the following observation equations 7

A = 68° 12' 36"

B = 53° 46' 12"

C = 58° 01' 16"

(d) The following round of angles was observed from central station to the surrounding stations of a triangulation survey

A = 93° 43' 22" wt – 3

B = 74° 32' 39" wt – 2

C = 101° 13' 44" wt – 2

D = 90° 29' 50" wt – 3

In addition one angle $(\overline{A+B})$ was measured seperately as combined angle with a mean value of 168° 16' 06" (wt – 2). Determine the most probable value of the angles A, B, C and D. 7

Unit-3

Q. 3. (a) What is the object of techeometry ? 2

(b) Derive the formulae for horizontal distance and vertical distance in tacheometeric surveying with staff vertical, sight inclined. 7

(c) Determine the gradiant from a point 'A' to a point 'B' from the following observations made with a tacheometre fitted with an anallictic lens. The constant of the instrument was 100 and the staff was held vertically. 7

Inst st^n	Staff point	Bearing	Vertical angle	Staff readings
P	A	134°	+10° 32'	1.360, 1.915, 2.470
	B	224°	+5° 6'	1.065, 1.885, 2.705

(d) Write short notes on : 7

(i) Subtence method

(ii) Tangential method

Unit-4

Q. 4. (a) Define "Principle point" of photograph. 2

(b) Discuss about the comparison between air photograph and map. 7

(c) Derive expression for various scales in a vertical photograph. 7

(d) Define and explain with neat sketches : 7

(i) Isocentre

(ii) Tilt

(iii) Swing

Unit-5

Q. 5. (a) What are the uses of hydrographic surveying ? 2

(b) What are the various method of locating sounding, describe any three ? 7

(c) Write short notes on : 7

(i) Shore line survey

(ii) Sounding gauges

(iii) Problems in hydrographic surveying.

(d) A, B and C are three visible stations in a hydrographical survey. The computed sides

of the triangle ABC are AB = 1130 m ; BC = 1372 m ; CA = 1889 m ; Out side this triangle P is established and its position is to be found by three point resection on A, B and C, the angles APB and BPC being respectively 42° 35' and 54° 20'. Determine the distances PA and PC. 7